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CHEMISTRY

TARGET : JEE Advanced – 2023

CAPS – 24

s-Block Elements and hydrogen & p block (group 13-14)

1. Which of the following compounds is consumed during the preparation of Na_2CO_3 by Solvay's process?

(A) NH_3, $CaCO_3$, NaCl (B) NH_4Cl, CaO, NaCl

(C) $CaCO_3$, NaCl (D) NaCl, NH_4HCO_3

2. II A (alkaline earth metals) and II B (zinc family) resemble

(A) $MgSO_4.7H_2O$ is isomorphous with $ZnSO_4.7H_2O$

(B) II A and II B cations are not precipitated by H_2S in acidic medium

(C) both (a) and (b)

(D) none of the above

3. $$KO_2 + CO_2 + H_2O \xrightarrow{\text{more } CO_2} [X] + [Y]$$

Products [X] and [Y] are respectively:

(A) K_2CO_3, O_2 (B) $KHCO_3$, O_2 (C) KOH, K_2CO_3 (D) $KHCO_3$, H_2O

4. Consider the following statements for diborane:

1. Boron is approximately sp3 hybridized

2. B–H–B angle is 180o

3. There are two terminal B–H bonds for each boron atom

4. There are only 12 bonding electrons available

Of these statements:

(A) 1, 3 and 4 are correct (B) 1, 2 and 3 are correct

(C) 2, 3 and 4 are correct (D) 1, 2 and 4 are correct

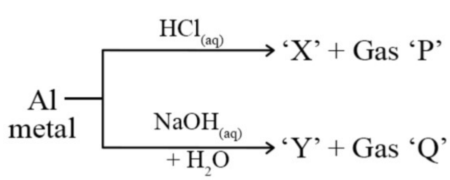
5. Borax is converted into amorphous boron by the following steps:

$$\text{Borax} \xrightarrow{X+H_2O} H_3BO_3 \xrightarrow{\Delta} B_2O_3 \xrightarrow{Y,\Delta} B$$

X and Y are respectively:

(A) HCl, Mg (B) HCl, C (C) C, Al (D) HCl, Al

6.



The incorrect statement regarding above reactions is:

(A) Al shows amphoteric character (B) Gas 'P' and 'Q' are different

(C) Both X and Y are water soluble (D) Gas Q is inflammable

7. The incorrect statement regarding 'X' in given reaction is:

$BF_3 + LiAlH_4 \xrightarrow{Ether} (X) + LiF + AlF_3$

(A) Twelve electrons are involved in bonding

(B) Four, two centre-two electron bonds

(C) Two, three centre-two electron bonds

(D) X does not react with NH_3

8. In which of the following silicates, only two corners per tetrahedron are shared?

(i) Pyro silicate (ii) Cyclic silicate

(iii) Double chain silicate (iv) Single chain silicate

(v) 3D silicate (vi) Sheet silicate

(A) (i), (ii) and (v) (B) (iv) and (vi) only

(C) (i) and (vi) only (D) (ii) and (iv) only

MULTIPLE CHOICE QUESTIONS

9. Which of the following will release CO_2 when heated to 1000ºC?

(A) $KHCO_3$ (B) Li_2CO_3 (C) K_2CO_3 (D) $PbCO_3$

10. Heating of oxalic acid with conc. H_2SO_4 evolves:

(A) CO (B) SO_2 (C) CO_2 (D) SO_3

COMPREHENSION # 1

$$\text{Metal} + \text{dil. HCl} \longrightarrow A \xrightarrow[NH_4OH]{Na_2HPO_4} B\ (\text{white ppt.})$$

$$A \xrightarrow{HCl(g)\ \text{Heated}} C$$

$$C \xrightarrow[\text{presence of NaCl}]{\text{Electrolysis in}} \text{Metal (M)}$$

11. The compound A is:

(A) $CaCl_2.2H_2O$ (B) $MgCl_2.6H_2O$ (C) $Na_2SO_4.10H_2O$ (D) $CaSO_4.2H_2O$

12. The compound B is:

(A) $Mg(NH_4)PO_4$ (B) $Ca_3(PO_4)_2 + NH_3$ (C) $Na(NH_4)HPO_4$ (D) both (a) and (b)

13. The compound C and metal M are:

(A) NaCl, Na (B) $CaCl_2$, Ca (C) $MgCl_2$, Mg (D) $BeCl_2$, Be

COMPREHENSION # 2

$A \xrightarrow{\Delta} B\ (\text{oxide}) + CO_2$

$B + H_2O \longrightarrow C$

$C + CO_2 \longrightarrow A\ (\text{milky})$

$C + NH_4Cl \xrightarrow{\Delta} D\ (\text{gas})$

$D + H_2O + CO_2 \longrightarrow E$

$E + NaCl \longrightarrow F$

$F \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

14. A is

(A) $Ca(HCO_3)_2$ (B) $CaCO_3$ (C) CaO (D) Na_2CO_3

15. B and C are

(A) CaO, $Ca(OH)_2$ (B) $Ca(OH)_2$, $CaCO_3$

(C) $CaCO_3$, $Ca(OH)_2$ (D) $Ca(OH)_2$, CaO

16. D, E and F are

(A) NH_3, NH_4Cl, NH_4HCO_3 (B) NH_3, NH_4HCO_3, $NaHCO_3$

(C) NH_4HCO_3, Na_2CO_3, $NaHCO_3$ (D) None

COMPREHENSION # 3

17. (i) $P + C(\text{carbon}) + Cl_2 \rightarrow Q + CO\uparrow$

(ii) $Q + H_2O \rightarrow R + HCl$

(iii) $BN + H_2O \rightarrow R + NH3\uparrow$

(iv) $Q + LiAlH_4 \rightarrow S + LiCl + AlCl_3$

(v) $S + H_2O \rightarrow R + H2\uparrow$

(vi) $S + NaH \rightarrow T$

(P, Q, R, S and T do not represent their chemical symbols)

Compound Q has:

I. zero dipole moment II. a planar trigonal structure

III. an electron deficient compound IV. a Lewis base

Choose the correct code:

(A) I, IV (B) I, II, IV (C) I, II, III (D) I, II, III, IV

18. (i) $P + C(\text{carbon}) + Cl_2 \rightarrow Q + CO\uparrow$ (ii) $Q + H_2O \rightarrow R + HCl$

(iii) $BN + H_2O \rightarrow R + NH_3\uparrow$ (iv) $Q + LiAlH_4 \rightarrow S + LiCl + AlCl_3$

(v) $S + H_2 \rightarrow R + H_2\uparrow$ (vi) $S + NaH \rightarrow T$

(P, Q, R, S and T do not represent their chemical symbols)

Compound T is used as a/an:

(A) oxidizing agent (B) complexing agent (C) bleaching agent (D) reducing agent

19. (i) $P + C(\text{carbon}) + Cl_2 \rightarrow Q + CO\uparrow$ (ii) $Q + H_2O \rightarrow R + HCl$

(iii) $BN + H_2O \rightarrow R + NH_3\uparrow$ (iv) $Q + LiAlH_4 \rightarrow S + LiCl + AlCl_3$

(v) $S + H_2 \rightarrow R + H_2\uparrow$ (vi) $S + NaH \rightarrow T$

(P, Q, R, S and T do not represent their chemical symbols)

Compound S is:

I. an odd-e^- compound

II. $(2c - 3e^-)$ compound

III. a electron deficient compound

IV. a sp^2 hybridized compounds.

Choose the correct code

(A) III (B) I, III (C) II, III, IV (D) I, II, IV

MATRIX TYPE QUESTIONS

20. Entries of Column-I are to be matched with entries of Column-II. Each entry of Column-I may have the matching with one or more than one entries of Column-II.

Column-I (Prop. of metals)	Column-II (Metals)
(A) Yellow flame colour	(P) Ca
(B) Most reactive with water	(Q) Mg
(C) Gives carbide with 'C'	(R) Na
(D) Metal nitrate $\xrightarrow{\Delta}$ metal oxide + NO_2 + O_2	(S) Li

(A) A $\rightarrow$ R; B $\rightarrow$ R; C $\rightarrow$ P, Q, S; D $\rightarrow$ P, Q, S

(B) A $\rightarrow$ P; B $\rightarrow$ Q; C $\rightarrow$ R, S; D $\rightarrow$ P, Q, S

(C) A $\rightarrow$ Q; B $\rightarrow$ P; C $\rightarrow$ S; D $\rightarrow$ Q, R, S

(D) A $\rightarrow$ S; B $\rightarrow$ P; C $\rightarrow$ Q, R; D $\rightarrow$ P, Q, R

SUBJECTIVE ANSWER TYPE

21. A certain salt 'X' gives the following test.

(i) Its aqueous solution is alkaline to litmus.

(ii) On strongly heating it swells to give glassy bead.

(iii) When concentrated H_2SO_4 is added to solution of 'X', white crystals of a weak acid. seperate out. Identify 'X' and write down all reaction.

22. A white crystalline comp. 'A' Swell up on heating and gives violet coloured flame on bunsen flame Its, aq. solution gives the following reaction.

(i) A white ppt. with $BaCl_2$ in presence of HCl.

(ii) When treated with excess of NH_4OH it gives white gelatinous ppt., which dissolve in NaOH.

23. A binary salt (A) on reaction with H_2O gives (B) aq. and (C) hydrocarbon. (C) gas on passing into ammonical $AgNO_3$ gives white ppt. (D). CO_2 gas turns (B) aq. milky. Identify (A), (B), (C) and (D).

24. Arrange the following in order of increasing ....

(i) Thermal stability	$BeSO_4$, $MgSO_4$, $CaSO_4$
(ii) Polarising power	Be^{2+}, Mg^{2+}, Ca^{2+}
(iii) Solubility in H_2O	$Be(OH)_2$, $Mg(OH)_2$, $Ca(OH)_2$
(iv) Covalent nature	$BeCl_2$, $MgCl_2$, $CaCl_2$
(v) Hydrolysis nature	$BeCl_2$ $MgCl_2$, $CaCl_2$
(vi) Lattice energy	CaF_2, MgF_2, BaF_2
(vii) Hydration energy	Be^{2+}, Mg^{2+}, Ba^{2+}
(viii) Basic nature	Be, Mg, Ca, Sr